

KC House Price Classification

Dataset Used: kc_house_data.csv

This project is based on the King County House Sales dataset (kc_house_data.csv) and focuses on applying machine learning classification algorithms using the Scikit-learn library. The main objective is to predict house price categories (low, medium, high) based on features such as bedrooms, bathrooms, square footage, condition, and grade.

The continuous price values are converted into categorical labels to make the problem suitable for classification. Multiple machine learning algorithms are applied and compared based on their accuracy, including Support Vector Machine, Logistic Regression, Naive Bayes, Decision Tree, and Random Forest.

The primary Scikit-learn modules used in this project are:

- **sklearn.svm:** For Support Vector Machine (SVC)
- **sklearn.linear_model:** For Logistic Regression
- **sklearn.naive_bayes:** For Naive Bayes classifier
- **sklearn.tree:** For Decision Tree classifier
- **sklearn.ensemble:** For Random Forest classifier
- **sklearn.model_selection:** For train-test splitting
- **sklearn.metrics:** For model evaluation using accuracy score

Project Objectives:

- To classify house prices into categories using machine learning algorithms.
- To implement multiple classification models using Scikit-learn.
- To compare the performance of different algorithms using accuracy score.
- To understand the impact of different classifiers on the same dataset.

Machine Learning Algorithms:

1. Support Vector Machine (SVM)

Definition:

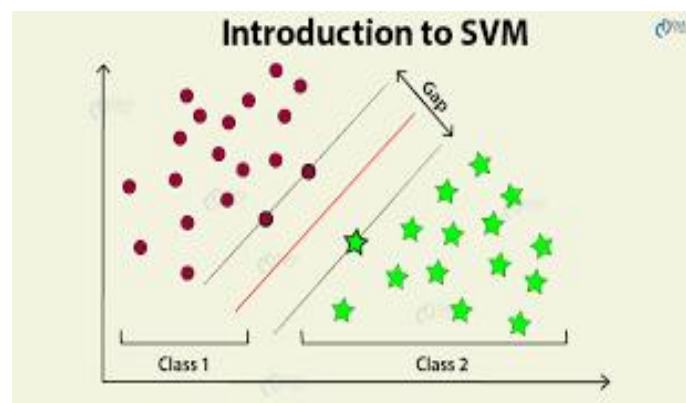
Support Vector Machine (SVM) is a supervised machine learning algorithm used for classification. It finds the optimal boundary (hyperplane) that separates different classes with the maximum margin.

Use

- Used for classification problems with clear margins
- Works well in high-dimensional data
- Effective when classes are separable

Key Point

- Uses distance → requires feature scaling



2. Logistic Regression

Definition

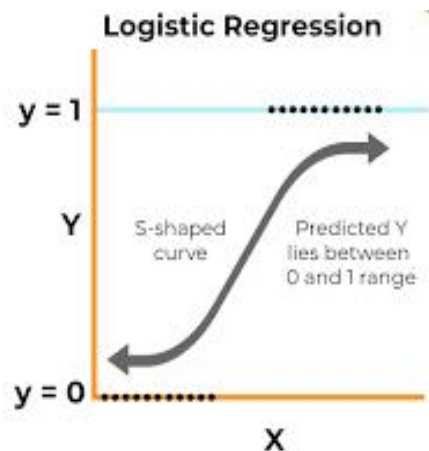
Logistic Regression is a classification algorithm that predicts the probability of a data point belonging to a particular class using a logistic (sigmoid) function.

Use

- Binary and multi-class classification
- Simple and fast model
- Works well with linearly separable data

Key Point

- Outputs probability, then converts to class .



3. Naive Bayes

Definition

Naive Bayes is a probabilistic classifier based on Bayes' Theorem, assuming that all features are independent of each other.

Use

- Text classification (spam detection, sentiment analysis)
- Works well with large datasets
- Fast and efficient

Key Point

- Assumes feature independence (not always realistic)

3. Decision Tree

Definition

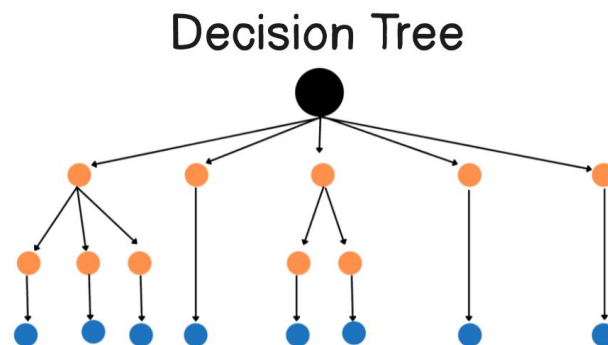
Decision Tree is a supervised learning algorithm that splits data into branches based on feature values, forming a tree-like structure for decision making.

Use

- Easy to understand and interpret
- Handles both classification and regression
- Works well with non-linear data

Key Point

- Can overfit easily



5. Random Forest

Definition

Random Forest is an ensemble learning method that builds multiple decision trees and combines their outputs to improve accuracy and reduce overfitting.

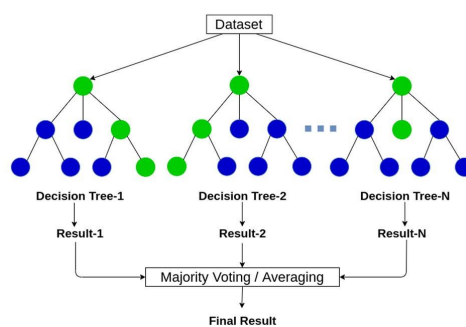
Use

- High accuracy models
- Handles large datasets
- Reduces overfitting compared to decision trees
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Key Point

- Uses multiple trees (bagging technique)

Random Forest



6. K-Nearest Neighbors (KNN)

Definition

KNN is a lazy learning algorithm that classifies data points based on the majority class of their nearest neighbors.

Use

- Simple and easy to implement
- Used in recommendation systems
- Works well with small datasets

Key Point

- Uses distance → requires feature scaling

7. Gradient Boosting

Definition

Gradient Boosting is an ensemble technique that builds models sequentially, where each new model corrects the errors of the previous one.

Use

- High-performance models
- Used in competitions (like Kaggle)
- Works well with complex datasets

Key Point

- Reduces errors step by step

Conclusion

Different machine learning algorithms were applied to classify house prices into categories. Tree-based models like Random Forest performed well, while KNN and SVM required feature scaling for better results. Logistic Regression and Naive Bayes provided simple probabilistic approaches. Overall, the study helped understand how different algorithms perform on the same dataset.

What You Have learned

- I learned how to apply different machine learning algorithms to a dataset and compare their performance.
- I understood the importance of data preprocessing, including feature selection and train-test splitting.
- I learned why feature scaling is necessary for algorithms like KNN and SVM.

Overall, I gained practical experience in building and evaluating classification models.